Mohamed Fedi BOUABID

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EDUCATION

National Institution of Applied Sciences and Technology **Sep. 2020 - Sep. 2025**
Engineering degree, Data science and intelligent systems *Tunis, Tunisia*

- **Relevant coursework :** Algebra, Calculus, Probability & Statistics, Algorithm, machine & deep learning.

WORK EXPERIENCE

Euklydia **Nov. 2025 – Present**
AI Engineer *Tunis, Tunisia*

- Architecting autonomous AI agents using **LangGraph** and **n8n** to automate marketing workflows and social media sentiment analysis.
- Training and deploying AI models into production via **FastAPI** and **Docker**, managing the full development lifecycle with **Git**.
    - **Stack : LangGraph, n8n, FastAPI, Docker, Git.**

EFREI Research Lab **Mar. 2025 – Aug. 2025**
NLP and Graph R&D Intern *Villejuif, France*

- Engineered a data pipeline to process **1M+ social media posts** into structured conversation trees and user interaction graphs using **NetworkX**.
- Fine-tuned a **BERTweetFr** model for domain-specific text classification, achieving a **94.2% F1 score**.
- Synthesized NLP outputs with **graph motifs** to analyze the correlation between sentiment, topic relevance, and user engagement patterns.
    - **Stack: NetworkX, PyTorch, HuggingFace, Transformers, Cytoscape, BERT, Gemini.**

E-Direct Tunisie **Jul. 2024 – Aug. 2024**
AI Developer Intern *Tunis, Tunisia*

- Designed and developed a web platform for automated recruitment.
- Integrated AI models for extracting, analyzing, and evaluating applications.
- Wrote comprehensive technical documentation to ensure the solution's maintainability and scalability
    - **Stack : Angular, NestJS, Flask, MySQL, YOLOv8, Gemini 2.0, OCR, LaTeX.**

PROJECTS

Job Email Classifier Dashboard | Github repository **2025**

- Streamlit dashboard that integrates Gmail and Gemini AI to automatically classify and visualize job-related emails with interactive analytics helping job seekers manage email responses.
    - **Stack : Langchain, Gemini API, Gmail API, Streamlit.**

NutriWise **2024**

- Mobile App that uses ML models hosted on Azure ML platform to assess food healthiness based on nutritional values retrieved from barcodes.
    - **Stack : Flutter, Firebase, Sci-kit learn , google ML Kit, Azure machine learning, Flask.**

RAG Project | Github repository **2024**

- Streamlit-based application that enables conversational querying of multiple PDF documents with natural language and interactive chat history.
    - **Stack : Langchain, ChromaDB, Llama 3.1, nomic-embed-text, Streamlit.**

MLOps Project | Github repository **2024**

- Hosted an XGBoost model on a Flask server within Docker containers, implemented monitoring with Prometheus and Grafana, and migrated the entire setup to Azure Machine Learning for scalable deployment.
    - **Stack : Sci-kit learn, Prometheus, Grafana, Docker, Azure ML.**

SKILLS

- **AI & Data Science :** NLP (Transformers, Hugging Face, PyTorch) · Topic Modeling (LDA, BERTopic) · Computer Vision (OCR, Vision Transformers) · RAG & AI Agents · Scikit-Learn · Data Analysis (Pandas)
- **Full-Stack :** Angular · React · NestJS · Flask · Django · JWT · WebSocket · MySQL · PostgreSQL
- **Cloud & Tools :** Firebase · Azure · Streamlit · Flutter · Power BI · Git · LaTeX
- **Languages :** French (Advanced Intermediate B2), English (Advanced Intermediate B2), Arabic (Native)